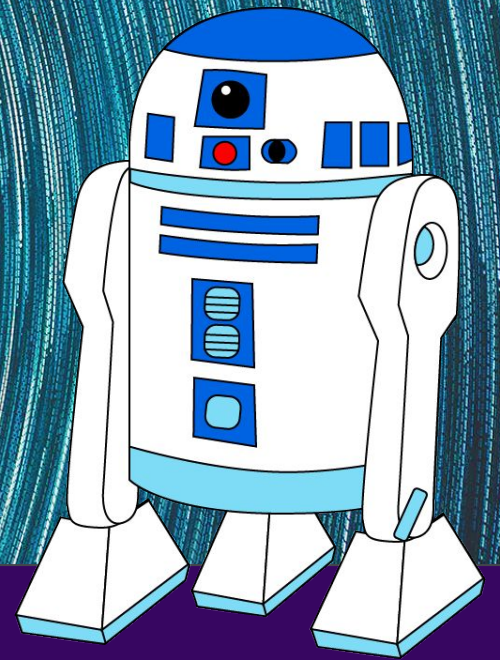


CIS 421/521:
ARTIFICIAL INTELLIGENCE

Probabilities and Language Models



Uncertainty

General situation:

- **Observed variables (evidence):** Agent knows certain things about the state of the world (e.g., sensor readings or symptoms)
- **Unobserved variables (states):** Agent needs to reason about other aspects (e.g. where an object is or what disease is present)
- **Model:** Agent knows something about how the known variables relate to the unknown variables

Probabilistic reasoning gives us a framework for managing our beliefs and knowledge



What Probabilities Are About

Like logical assertions, probabilities are about **possible worlds**. Instead of strictly ruling out possibilities (where a logical assertion is false), probabilities quantify **how likely** a particular possible world is.

In probability theory, the possible worlds are called the **sample space**, and they **mutually exclusive** and **exhaustive**.

A fully specified probability model associates a probability $P(w)$ with each possible world w .

Random Variables

A random variable is some aspect of the world about which we (may) have uncertainty

- R = Is it raining?
- U = Is the professor carrying an umbrella?

We denote random variables with capital letters



Axioms of Probability

The probability of any possible world is between 0 and 1.

$$0 \leq P(w) \leq 1 \text{ for every } w$$

The total probability of the set of all possible worlds is 1:

$$\sum_{w \in \Omega} P(w) = 1$$

Probability Distributions

Unobserved random variables have distributions

$P(T)$		$P(W)$	
T	P	W	P
hot	0.5	sun	0.6
cold	0.5	rain	0.1
		fog	0.3

A distribution is a TABLE of probabilities of values

A probability (lower case value) is a single number

Must have: $P(W = \text{rain}) = 0.1$ and

$$\forall x \ P(X = x) \geq 0$$

Shorthand notation:

$$P(\text{hot}) = P(T = \text{hot}),$$

$$P(\text{cold}) = P(T = \text{cold}),$$

$$P(\text{rain}) = P(W = \text{rain}),$$

...

OK if all domain entries are unique

$$\sum_x P(X = x) = 1$$

Joint Distributions

A *joint distribution* over a set of random variables: $X_1, X_2, \dots, X_n$ specifies a real number for each assignment (or *outcome*):

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey:

$$P(x_1, x_2, \dots, x_n) \geq 0$$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Size of distribution if n variables with domain sizes d?

- For all but the smallest distributions, impractical to write out!

Probabilistic Models

A probabilistic model is a joint distribution over a set of random variables

Probabilistic models:

- (Random) variables with domains
- Assignments are called *outcomes*
- Joint distributions: say whether assignments (outcomes) are likely
- *Normalized*: sum to 1.0
- Ideally: only certain variables directly interact

Distribution over T,W

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Events

An *event* is a set E of outcomes

$$P(E) = \sum_{(x_1 \dots x_n) \in E} P(x_1 \dots x_n)$$

From a joint distribution, we can calculate the probability of any event

- Probability that it's hot AND sunny?
- Probability that it's hot?
- Probability that it's hot OR sunny?

Typically, the events we care about are *partial assignments*, like $P(T=\text{hot})$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Marginal Distributions

Marginal distributions are sub-tables which eliminate variables

Marginalization (summing out): Combine collapsed rows by adding

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



$$P(t) = \sum_s P(t, s)$$

$P(T)$

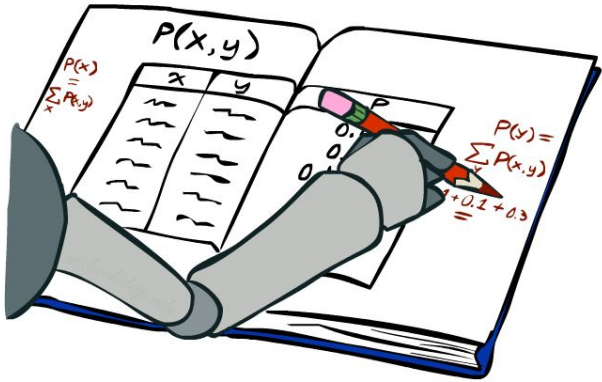
T	P
hot	0.5
cold	0.5



$$P(s) = \sum_t P(t, s)$$

$P(W)$

W	P
sun	0.6
rain	0.4



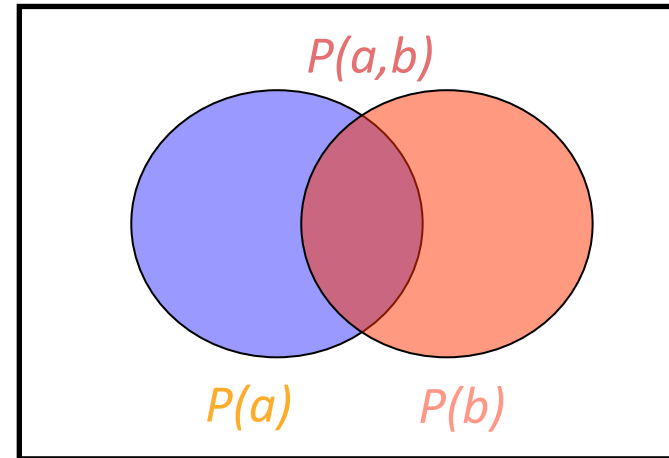
$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Conditional Probabilities

A simple relation between joint and conditional probabilities

- In fact, this is taken as the *definition* of a conditional probability

$$P(a|b) = \frac{P(a, b)}{P(b)}$$



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{0.2}{0.5} = 0.4$$

$$\begin{aligned} &= P(W = s, T = c) + P(W = r, T = c) \\ &= 0.2 + 0.3 = 0.5 \end{aligned}$$

Conditional Distributions

Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

$P(W T)$	$P(W T = \text{hot})$	
	W	P
	sun	0.8
	rain	0.2
	$P(W T = \text{cold})$	
	W	P
	sun	0.4
	rain	0.6

Joint Distribution

$P(T, W)$		
T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Probabilistic Inference

Probabilistic inference: compute a desired probability from other known probabilities (e.g. conditional from joint)

We generally compute conditional probabilities

- $P(\text{on time} \mid \text{no reported accidents}) = 0.90$
- These represent the agent's *beliefs* given the evidence

Probabilities change with new evidence:

- $P(\text{on time} \mid \text{no accidents, 5 a.m.}) = 0.95$
- $P(\text{on time} \mid \text{no accidents, 5 a.m., raining}) = 0.80$
- Observing new evidence causes *beliefs to be updated*



Inference by Enumeration

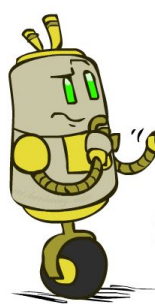
General case:

- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query* variable: Q
 - Hidden variables: $H_1 \dots H_r$
- $$\left. \begin{array}{l} E_1 \dots E_k = e_1 \dots e_k \\ Q \\ H_1 \dots H_r \end{array} \right\} \begin{array}{l} X_1, X_2, \dots X_n \\ \text{All variables} \end{array}$$

■ We want:

$$P(Q|e_1 \dots e_k)$$

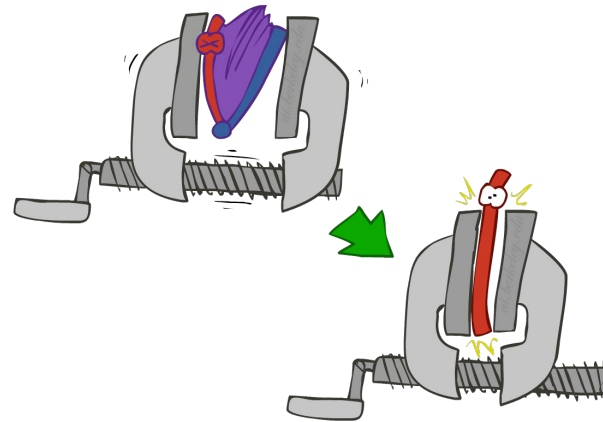
■ Step 1: Select the entries consistent with the evidence



x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

2 0.15

■ Step 2: Sum out H to get joint of Query and evidence



■ Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$

$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} P(Q, \underbrace{h_1 \dots h_r}_{X_1, X_2, \dots X_n}, e_1 \dots e_k)$$

The Product Rule

$$P(y)P(x|y) = P(x, y)$$

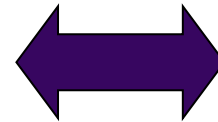
Example:

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3



$P(D, W)$

D	W	P
wet	sun	
dry	sun	
wet	rain	
dry	rain	

The Chain Rule

More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots, x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$

Why is this always true?

Bayes' Rule

Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

Dividing, we get:

$$P(x|y) = \frac{P(y|x)}{P(y)}P(x)$$

Why is this at all helpful?

- Lets us build one conditional from its reverse
- Often one conditional is tricky but the other one is simple
- Foundation of many systems we'll see later (e.g. ASR, MT)

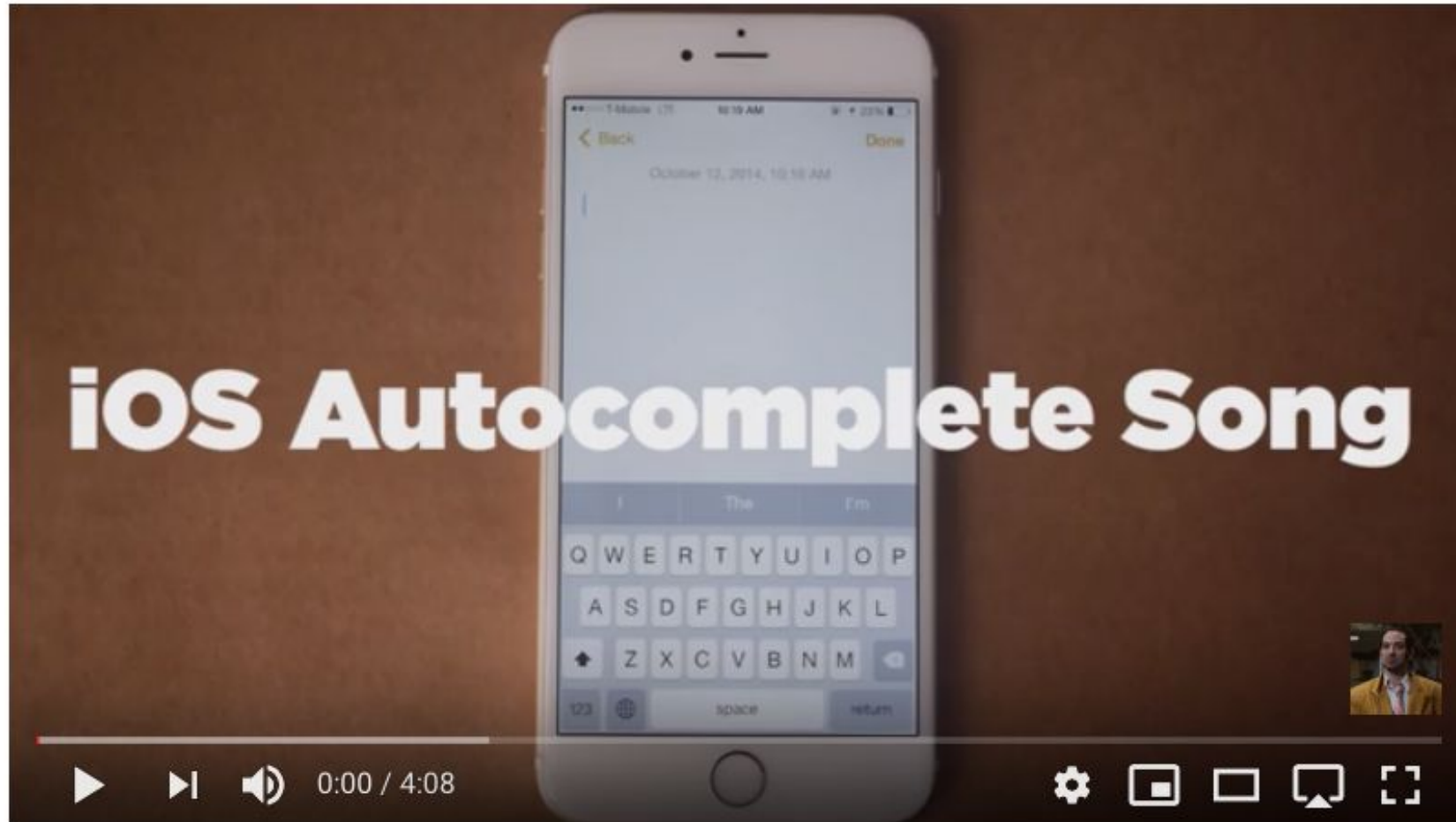
In the running for most important AI equation!





YouTube

Search



🎵 iOS Autocomplete Song | Song A Day #2110

<https://www.youtube.com/watch?v=M8MJFrdfGe0>

Probabilistic Language Models

Probabilistic Language Models

One goal: assign a probability to a sentence

- Autocomplete for texting
- Machine Translation
- Spelling Correction
- Speech Recognition

Other Natural Language Generation tasks: summarization, question-answering, dialog systems

Probabilistic Language Modeling

Goal: compute the probability of a sentence or sequence of words

Related task: probability of an upcoming word

A model that computes either of these is called a **language model** or **LM**

Probabilistic Language Modeling

Goal: compute the probability of a sentence or sequence of words

$$P(W) = P(w_1, w_2, w_3, w_4, w_5 \dots w_n)$$

Related task: probability of an upcoming word

$$P(w_5 | w_1, w_2, w_3, w_4)$$

A model that computes either of these

$P(W)$ or $P(w_n | w_1, w_2 \dots w_{n-1})$ is called a **language model**.

Better: **the grammar** But **language model** or **LM** is standard

How to compute $P(W)$

How to compute this joint probability:

$P(\text{the, underdog, Philadelphia, Eagles, won})$

Intuition: let's rely on the Chain Rule of Probability

The Chain Rule



The Chain Rule

Recall the definition of conditional probabilities

$$p(\mathbf{B}|\mathbf{A}) = \mathbf{P}(\mathbf{A},\mathbf{B})/\mathbf{P}(\mathbf{A}) \quad \text{Rewriting: } \mathbf{P}(\mathbf{A},\mathbf{B}) = \mathbf{P}(\mathbf{A})\mathbf{P}(\mathbf{B}|\mathbf{A})$$

More variables:

$$P(A,B,C,D) = P(A)P(B|A)P(C|A,B)P(D|A,B,C)$$

The Chain Rule in General

$$P(x_1, x_2, x_3, \dots, x_n) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2) \dots P(x_n|x_1, \dots, x_{n-1})$$

Joint probability of words in sentence

Joint probability of words in sentence

$$P(w_1 w_2 \dots w_n) = \prod_i P(w_i \mid w_1 w_2 \dots w_{i-1})$$

$P(\text{"the underdog Philadelphia Eagles won"}) =$

$P(\text{the}) \times$
 $P(\text{underdog} \mid \text{the}) \times$
 $P(\text{Philadelphia} \mid \text{the underdog}) \times$
 $P(\text{Eagles} \mid \text{the underdog Philadelphia}) \times$
 $P(\text{won} \mid \text{the underdog Philadelphia Eagles})$

How to estimate these probabilities

Could we just count and divide?

How to estimate these probabilities

Could we just count and divide? Maximum likelihood estimation (MLE)

$$P(\text{won} \mid \text{the underdog team}) = \frac{\text{Count}(\text{the underdog team won})}{\text{Count}(\text{the underdog team})}$$

Why doesn't this work?

Simplifying Assumption = Markov Assumption

Simplifying Assumption = Markov Assumption

$P(\text{won} | \text{the underdog team}) \approx P(\text{won} | \text{team})$

Only depends on the previous k words, not the whole context

$\approx P(\text{won} | \text{underdog team})$

$\approx P(w_i | w_{i-2} w_{i-1})$

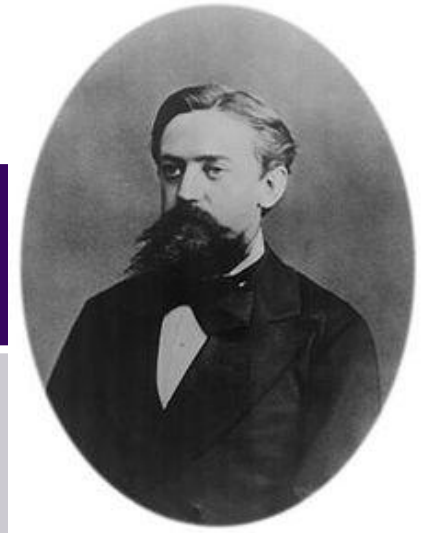
$$P(w_1 w_2 w_3 w_4 \dots w_n) \approx \prod_i^n P(w_i | w_{i-k} \dots w_{i-1})$$

K is the number of context words that we take into account

How much history should we use?

unigram	no history		
bigram	1 word as history		
trigram	2 words as history		
4-gram	3 words as history		

Historical Notes



1913	Andrei Markov counts 20k letters in <i>Eugene Onegin</i>
1948	Claude Shannon uses n-grams to approximate English
1956	Noam Chomsky decries finite-state Markov Models
1980s	Fred Jelinek at IBM TJ Watson uses n-grams for ASR, think about 2 other ideas for models: (1) MT, (2) stock market prediction
1993	
	Jelinek left IBM to found CLSP at JHU Peter Brown and Robert Mercer move to Renaissance Technology

Simplest case: Unigram model

$$P(w_1 w_2 \dots w_n) \approx \prod_i P(w_i)$$

Some automatically generated sentences from a unigram model

fifth an of futures the an incorporated a a the inflation most
dollars quarter in is mass

thrift did eighty said hard 'm july bullish

that or limited the

Bigram model

- Condition on the previous word:

$$P(w_i \mid w_1 w_2 \dots w_{i-1}) \approx P(w_i \mid w_{i-1})$$

texaco rose one in this issue is pursuing growth in a boiler house said
mr. gurria mexico 's motion control proposal

without permission from five hundred fifty five yen

outside new car parking lot of the agreement reached

this would be a record november

N-gram models

We can extend to trigrams, 4-grams, 5-grams

In general this is an insufficient model of language

- because language has **long-distance dependencies**:

“The computer(s) which I had just put into the machine room on the fifth floor is (are) crashing.”

But we can often get away with N-gram models

Language Modeling

Estimating N-gram Probabilities

Estimating bigram probabilities

The Maximum Likelihood Estimate

$$P(w_i \mid w_{i-1}) = \frac{\textit{count}(w_{i-1}, w_i)}{\textit{count}(w_{i-1})}$$

$$P(w_i \mid w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

An example

$$P(w_i | w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

<s> I am Sam </s>

<s> Sam I am </s>

<s> I do not like green eggs and ham </s>

$$P(\text{I} | \text{<s>}) = \frac{2}{3} = .67$$

$$P(\text{Sam} | \text{<s>}) = \frac{1}{3} = .33$$

$$P(\text{am} | \text{I}) = \frac{2}{3} = .67$$

$$P(\text{</s>} | \text{Sam}) = \frac{1}{2} = 0.5$$

$$P(\text{Sam} | \text{am}) = \frac{1}{2} = .5$$

$$P(\text{do} | \text{I}) = \frac{1}{3} = .33$$

Problems for MLE

Zeros

Train	Test
denied the allegations	denied the memo
denied the reports	
denied the claims	
denied the requests	

$P(\text{memo} | \text{denied the}) = 0$

And we also assign 0 probability to all sentences containing it!

Problems for MLE

Out of vocabulary items (OOV)

<unk> to deal with OOVs

Fixed lexicon L of size V

Normalize training data by replacing any word not in L with <unk>

Avoid zeros with smoothing

Practical Issues

We do everything in log space

- Avoid underflow
- (also adding is faster than multiplying)

$$\log(p_1 \times p_2 \times p_3 \times p_4) = \log p_1 + \log p_2 + \log p_3 + \log p_4$$

Google N-Gram Release, August 2006

AUG

3

All Our N-gram are Belong to You

Posted by Alex Franz and Thorsten Brants, Google Machine Translation Team

Here at Google Research we have been using word [n-gram models](#) for a variety of R&D projects,

...

That's why we decided to share this enormous dataset with everyone. We processed 1,024,908,267,229 words of running text and are publishing the counts for all 1,176,470,663 five-word sequences that appear at least 40 times. There are 13,588,391 unique words, after discarding words that appear less than 200 times.

Google N-Gram Release

serve as the incoming 92

serve as the incubator 99

serve as the independent 794

serve as the index 223

serve as the indication 72

serve as the indicator 120

serve as the indicators 45

serve as the indispensable 111

serve as the indispensable 40

serve as the individual 234

<http://googleresearch.blogspot.com/2006/08/all-our-n-gram-are-belong-to-you.html>